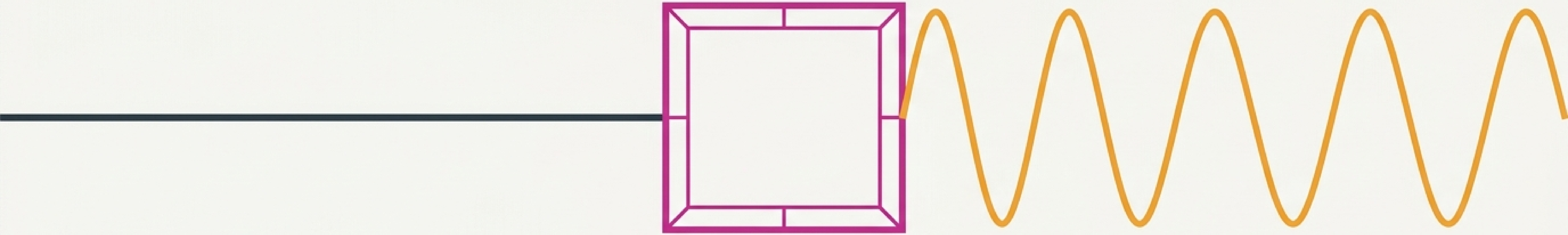
Survey of Quantum Technologies
& Applications (PH 40001)



Single Qubit Quantum Gates

Mastering the Quantum Control Panel

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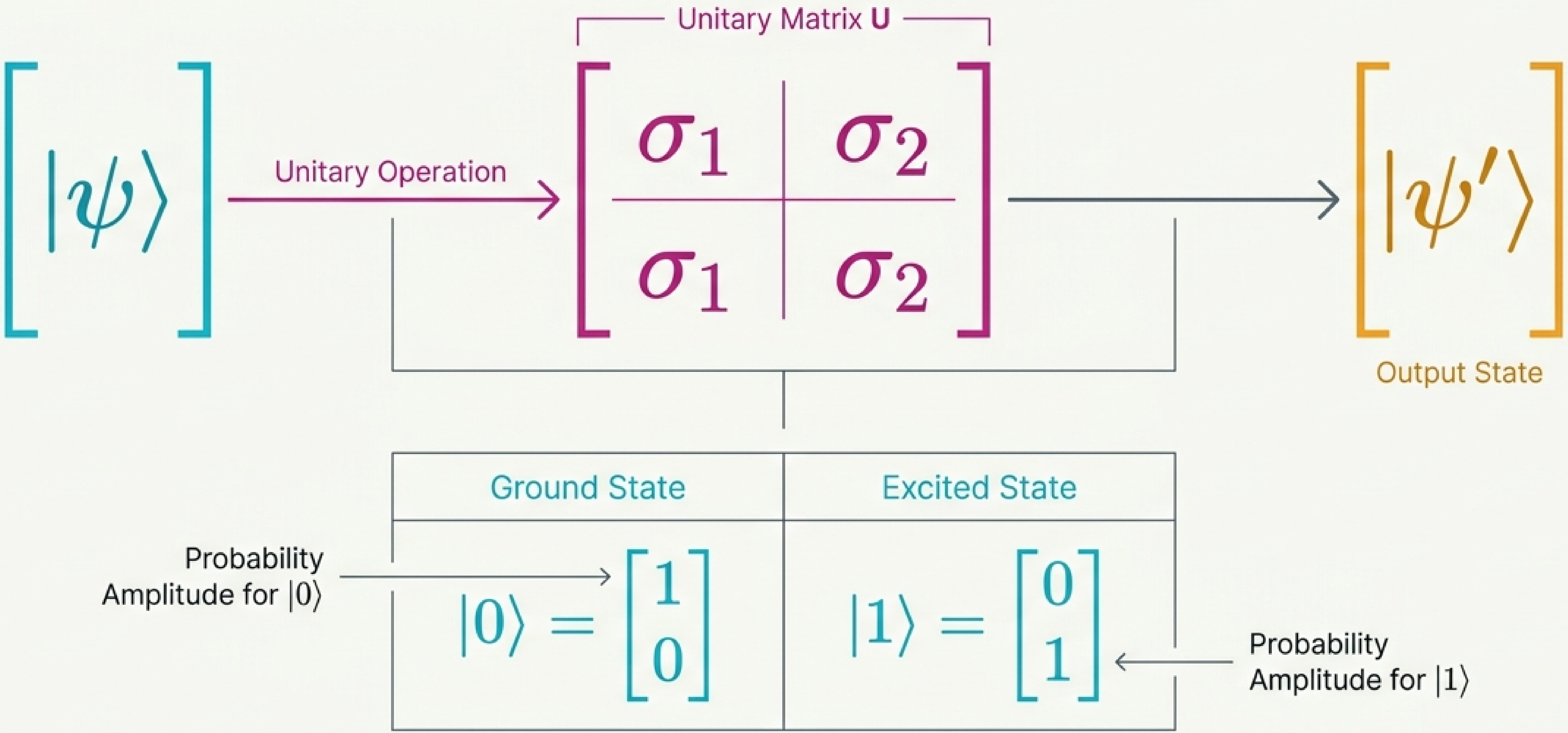
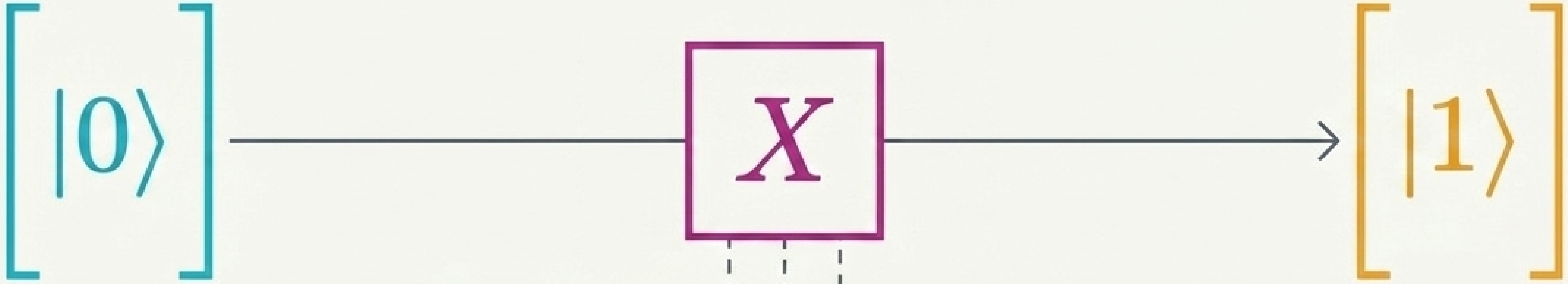
The Core Framework: Vectors & Matrices

Quantum states ($|\psi\rangle$) are represented as 2D column vectors. Single-qubit quantum gates are 2x2 unitary matrices that multiply these vectors.



Pauli X Gate: The Quantum Bit Flip

Acts exactly like a classical NOT gate.
It flips a $|0\rangle$ to a $|1\rangle$ (and vice versa).



$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \times \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Pauli Z Gate: The Phase Flip

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

A **purely quantum operation.** Leaves the ground state alone, but flips the phase (adds a negative sign) of the excited state.

Action on Ground State

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

Output is unchanged: $|0\rangle$

Action on Excited State

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix} = -|1\rangle$$

Phase Change

The Hadamard Gate (H): Entering Superposition

The H gate is the gateway to quantum power. It takes a definitive ground state and splits it into an equal superposition of both $|0\rangle$ and $|1\rangle$.

$$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix}1 & 1\\ 1 & -1\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix}1\\1\end{bmatrix} = \frac{1}{\sqrt{2}}\left(\begin{bmatrix}1\\0\end{bmatrix} + \begin{bmatrix}0\\1\end{bmatrix}\right)$$

$$\downarrow$$

Equal Superposition

$$|\psi\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$$

Hadamard on the Excited State: Creating the Phase Difference

When applied to $|1\rangle$, the H gate still creates an equal superposition, but the initial state dictates the phase. The resulting state features a critical phase flip.

gate still creates an equal superposition, but the initial state dictates the phase. The resulting state features a critical phase flip.

$$H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$= \frac{1}{\sqrt{2}} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} - \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right)$$

$$\downarrow$$

Critical Phase Difference

$$|\psi\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle - |1\rangle \right)$$

Circuit Walkthrough: Combining Gates

Quantum algorithms are built by chaining unitary operations. Here, a bit flip primes the state before the Hadamard gate pushes it into a phase-flipped superposition.

$$|0\rangle \quad \rightarrow \quad \rightarrow \quad |1\rangle \quad \rightarrow \quad \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$$

Continuous Control: Rotation Gates

Unlike Pauli gates which perform fixed 180-degree flips, Rotation Gates allow continuous, arbitrary angle shifts (θ) along the x, y, or z axes.

X-Axis Rotation	Y-Axis Rotation	Z-Axis Rotation
$R_x(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -i\sin\left(\frac{\theta}{2}\right) \\ -i\sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$	$R_y(\theta) = \begin{bmatrix} \cos\left(\frac{\theta}{2}\right) & -\sin\left(\frac{\theta}{2}\right) \\ \sin\left(\frac{\theta}{2}\right) & \cos\left(\frac{\theta}{2}\right) \end{bmatrix}$	$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$

$$|0\rangle \longrightarrow \boxed{R_n(\theta)} \longrightarrow |\psi\rangle$$

Phase Gate (S)

$$S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

90-degree phase shift

π/8 Gate (T)

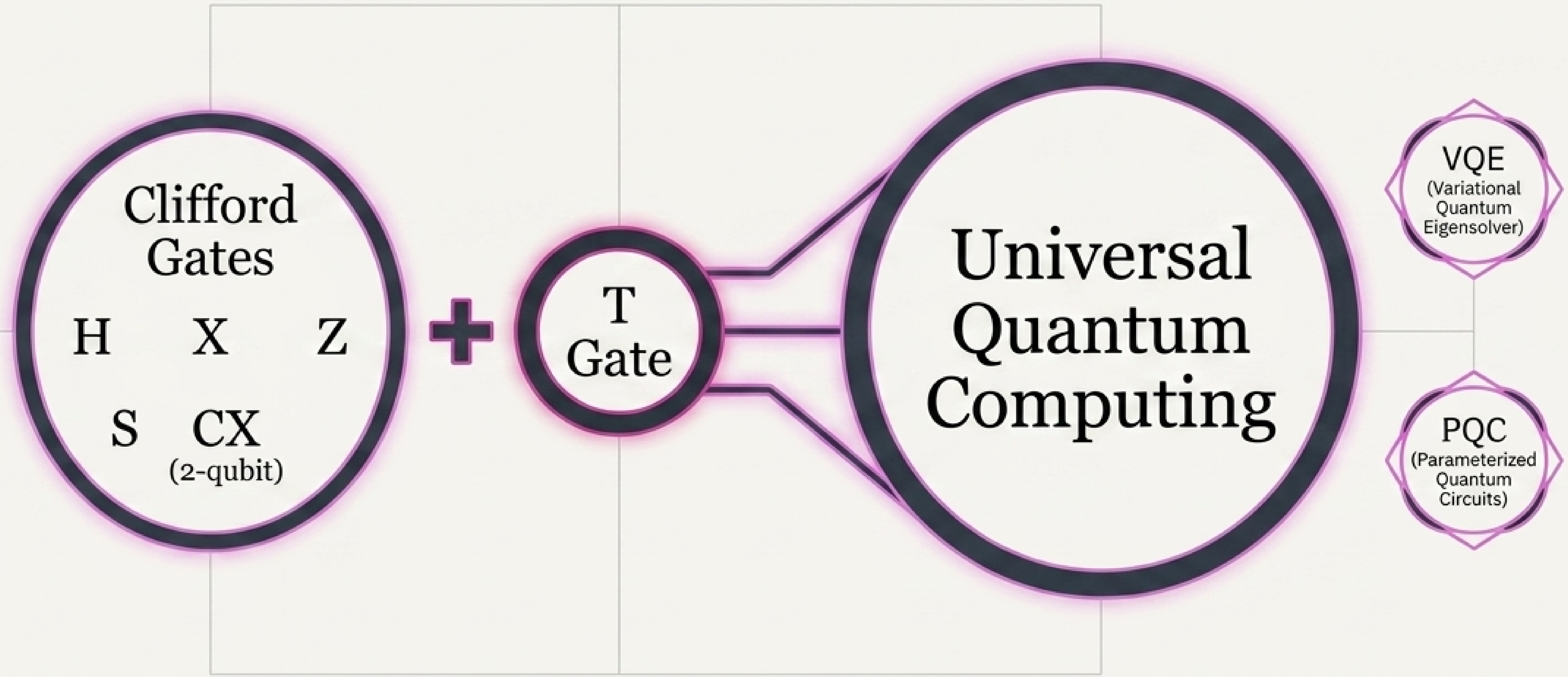
$$T = \begin{bmatrix} 1 & 0 \\ 0 & \boxed{e^{i\pi/4}} \end{bmatrix}$$

45-degree phase shift

$$= e^{i\pi/8} \begin{bmatrix} e^{-i\pi/8} & 0 \\ 0 & e^{i\pi/8} \end{bmatrix}$$

The Path to Universality: Clifford + T

A finite set of gates can approximate any continuous quantum operation. The "Clifford + T" combination is the standard recipe for Universal Quantum Computing. Used in algorithms like VQE and PQC.



Single-Qubit Gate Matrix Reference

Gate Name	Symbol	Matrix Structure	Action on $\|0\rangle$	Action on $\|1\rangle$
Pauli X (Bit Flip)	$\boxed{X}$	$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$	$\|1\rangle$	$\|0\rangle$
Pauli Z (Phase Flip)	$\boxed{Z}$	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	$\|0\rangle$	$-\|1\rangle$
Hadamard (Superposition)	$\boxed{H}$	$\frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$	$\frac{1}{\sqrt{2}}(\|0\rangle + \|1\rangle)$	$\frac{1}{\sqrt{2}}(\|0\rangle - \|1\rangle)$
Phase (S)	$\boxed{S}$	$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$	$\|0\rangle$	$i\|1\rangle$
$\frac{\pi}{8}\,(T)$	$\boxed{T}$	$\begin{bmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{bmatrix}$	$\|0\rangle$	$e^{i\pi/4}\|1\rangle$